Constructing Cloud-based Cyber Threat Intelligence Sharing and Correlation System

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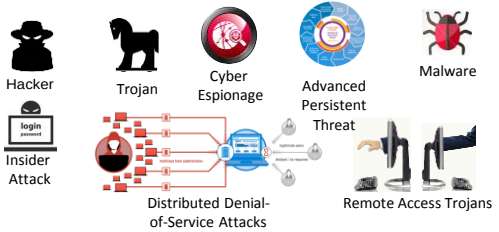
Introduction

A team of faculties and student at Oakland University is developing a cloud-based Cyber Threat Intelligence Sharing and Correlation System (CTISCS) against advanced cyber intrusions. The general goal of this project is to protect IT assets, critical infrastructure, and sensitive data against cyber intrusions by providing secure, effective, and affordable risk management services. Specifically, we are designing and implementing a cyber event correlation system and "Machine Learning-as-a-Service" (*MLaaS*) framework that aggregates, correlates, and disseminates cyber events in a free cloud infrastructure. The expected contributions of this project include

1) A testbed that investigates cyber threats,
2) An improved methodology that correlates cyber events, and
3) A viable solution that leverages the Cloud as the elastic backbone.

The successful deployment of the system will lead to a faster response and more accurate detection of sophisticated cyber intrusions launched on a wide scale, including the Advanced Persistent Threats (APTs), Remote Access Trojans (RATs), and Distributed Denial-of-Service (DDoS) Attacks.



System Design

Overview

- Construct three domains: ***Attack Domain***, ***Target Domain***, and ***Control Domain***.
- Each domain can properly be separated.
- The exploits can be launched, monitored, constrained, and studied later.

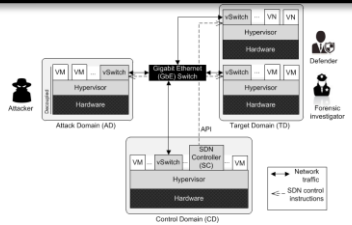


Fig. 1 The System Overview

Key Technologies

- Use **`CloudLab`** (https://www.cloudlab.us/) as the free cloud infrastructure.
- Use **`OpenStack`** (https://www.openstack.org/) to construct the virtual network.
- Use **`Python`** to profile the virtual network artifacts.

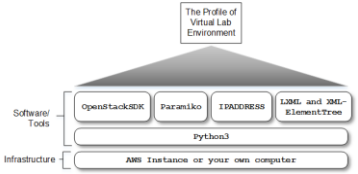


Fig. 2 The Key Technologies Used in This Project

Educational Components

Pervasive Adoption

- The cloud environment allows hands-on virtual cybersecurity labs to be hosted in a realistic setting, where the offensive, defensive, and forensic lab tasks can be performed
- Appealing advantages: ***free***, ***easy-to-setup***, ***scalable***, ***secure***, and ***easily-to-be-shared*** among institutions for different educational needs.
- The profile of virtual environment is saved in the .XML format.
  - Easy to be understood
  - Easy to be adopted and customized


```
<?xml version="1.0" ?>
<Profile>
  <Interfaces>
    <Interface ID="213fbc80-84c7-43e4-acc3-40de8fe434eb">
      <Interface_Name/>
      <Network_ID>5a742f66-0a82-498a-b5f1-5d02240750ec</Network_ID>
      <Interface_Admin_state_UP>True</Interface_Admin_state_UP>
      <Connected_Device_ID>2e5136f9-d4b7-432c-9a19-c882df8baade</Connected_Device_ID>
      <InterfaceSubnet_ID="12a8c86f-5b67-4981-b3de-ad91189cc856">
        <InterfaceIP>128.110.155.153</InterfaceIP>
      </InterfaceSubnet>
    </Interface>
...
```


Fig. 3 The Profile of Virtual Environment in .XML format

Our Team

- Three undergraduate students,
- Three master students
- One Ph.D. student at Oakland University
- Two faculty advisors

Tasks and Milestones

Current status

- Implemented the major components of the system.
- Implemented a library and a command-line interface.
- The virtual environment and VM instances can be profiled and deployed automatically.
- We are deploying advanced cyber attacks (e.g., Mirai botnet) on this environment.

Tab. 1 The Virtual Environment under Construction

Constructing Virtual Environment (Work in Progress)			
Sub-tasks	Required Software, Libraries, and Languages	Status	
		Prototype	Refinement
1: Setup the control domain (CD)	OpenStack Python API and AlienVault USM	✓	✓
2: Start/shutdown the VMs	OpenStack Python API	✓	✓
3: Profile the virtual environment of the labs	OpenStack Python API	✓	✓
4: Script the snapshotting and restoration of each domain	OpenStack Python API and Hypervisor-specific command and API	✓	✓
5: Change the topology of the overlay network	Open vSwitch (OVS) API	⚠	
6: Change routing rules	Open vSwitch (OVS) API	⚠	

Tab. 2 The Security-related Virtual Hands-on Labs

Security-related Virtual Labs (Work in Progress)		
Lab Description	Status	
	Prototype	Refinement
1: Cloud-based malware construction and virtual machine forensics	✓	✓
2: Fine-grained Access Control with Attribute-based Encryption (ABE)	✓	✓
3: Processing encrypted data with Homomorphic Encryption (HE)	✓	✓

Acknowledgement

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